

Sai Krishnam Raju Mudunuri

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 3+ years of experience building finance RAG applications and healthcare ML pipelines on AWS. Focused on retrieval quality, inference cost, and reliable model delivery.

WORK EXPERIENCE

AI/ML Engineer | Honeywell | Georgia

Oct 2024 - Present

- Reduced inference costs by **approximately 30%** by routing finance queries by complexity and sensitivity: approximately 65% went to open-source models on vLLM, while sensitive queries used Claude on Bedrock.
- Built finance RAG with Bedrock, Titan embeddings, and OpenSearch hybrid search (BM25 and vector); added metadata filters, reranking, and citations linking answers to source documents.
- Implemented LangGraph retrieval and tool-execution workflows with authenticated FastMCP access to internal APIs, scoped permissions, and output validation.
- Fine-tuned open-source LLMs with LoRA/QLoRA for financial-document understanding.
- Deployed quantized LLMs on GPU-enabled Amazon EKS with vLLM; configured continuous batching and tensor parallelism, using PagedAttention for KV-cache management.
- Evaluated RAG outputs for faithfulness and context precision/recall with RAGAS and DeepEval; used LangSmith to trace workflow execution and CloudWatch to monitor latency.
- Managed Terraform configurations and Helm charts for ML services; used Argo CD GitOps to synchronize Kubernetes deployments with version-controlled manifests.
- Built model release pipelines with SageMaker Pipelines, Model Registry, and GitHub Actions, using canary deployments and automated rollback to reduce production release risk.

ML Engineer | Infosys | India

Jan 2022 - Jul 2023

- Developed XGBoost, LightGBM, and Random Forest models for readmission risk and disease progression; used SHAP to review predictions with clinical stakeholders.
- Engineered clinical risk features from **15M+** EHR, lab, and patient-encounter records using Python and PySpark.
- Evaluated missed high-risk cases and false alerts using recall and precision; compared models with cross-validation, F1-score, and ROC-AUC, tracking experiments in MLflow.
- Reduced manual chart-review effort by **60%** with ClinicalBERT pipelines that extracted diagnoses, medications, and clinical observations from unstructured physician notes.
- Cut model deployment time by **50%** by automating deployment, versioning, and retraining on SageMaker with Docker, MLflow, and CI/CD workflows.
- Reduced model-quality incidents by **30%** by implementing drift monitoring with SageMaker Model Monitor, CloudWatch, and MLflow.

TECHNICAL SKILLS

Languages & ML Frameworks: Python, PyTorch, TensorFlow, SQL, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, PySpark

Generative AI & LLMs: Hugging Face, LangChain, LangGraph, MCP / FastMCP, LoRA/QLoRA, vLLM, ClinicalBERT

Vector Search & Retrieval: OpenSearch, FAISS, Pinecone, Redis

Cloud & Microservices: AWS (Bedrock, SageMaker, EKS, S3), FastAPI, REST APIs

MLOps & Infrastructure: Docker, Kubernetes, Terraform, Helm, Argo CD, GitHub Actions, Jenkins, Kubeflow, MLflow, DVC, Ansible

Evaluation & Observability: RAGAS, DeepEval, LangSmith, CloudWatch, SHAP

EDUCATION

Master of Science in Information System Technology | Wilmington University, Delaware

Bachelor of Technology in Computer Science and Engineering | Rajalakshmi Engineering College, Chennai, India